

Question	Marking details	Marks Available					
		AO1	AO2	AO3	Total	Maths	Prac
8	Indicative content Section A: From epidermis and across cortex: • Apoplast + symplast + vacuolar pathway • (Apoplast) Along cell walls • (Symplast) through cytoplasm/ plasmodesmata • (Vacuolar pathway) passes through tonoplast/ cytoplasm • Down water potential gradient / by osmosis Section B: Into vascular tissue: • Endodermis has Casparian strip • Waterproof / composed of suberin • Blocks apoplast pathway • Water {forced / passes} into symplast • Active transport of {salts / minerals} <u>into xylem</u> • Lowers water potential in <u>xylem</u> • Water enters <u>xylem</u> by osmosis / down water potential gradient Waterlogged soil: • {Less / no} oxygen (for respiration) • {Less / no} respiration so {less / no} {energy / ATP} • for active transport of minerals in the {endodermis/ root hairs} Must be in context of no oxygen • Lower water potential gradient as a result of less active transport of salts / minerals into xylem • Less water enters xylem which lowers root pressure	5	4				

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				7-9 marks Indicative content of this level is detailed content from all three areas <i>The candidate constructs an articulate, integrated account, correctly linking relevant points, such as those in the indicative content, which shows sequential reasoning. The answer fully addresses the question with no irrelevant inclusions or significant omissions. The candidate uses scientific conventions and vocabulary appropriately and accurately.</i> 4-6 marks Indicative content of this level is detailed content from two areas or less detail from three areas <i>The candidate constructs an account correctly linking some relevant points, such as those in the indicative content, showing some reasoning. The answer addresses the question with some omissions. The candidate usually uses scientific conventions and vocabulary appropriately and accurately.</i> 1-3 marks Indicative content of this level is any correct statement from indicative content <i>The candidate makes some relevant points, such as those in the indicative content, showing limited reasoning. The answer addresses the question with significant omissions. The candidate has limited use of scientific conventions and vocabulary.</i> 0 marks <i>The candidate does not make any attempt or give a relevant answer worthy of credit.</i>						
				Question 8 total	5	4	0	9	0	0